

ing employment. [Simon \(2025\)](#) documents that entry-level job postings have declined more than 35 percent since January 2023, with the steepest drops in highly exposed roles: a 10-point increase in exposure predicts an 11 percent decline in entry-level demand, while senior roles in those same occupations increase by 7 percent. [Dominski and Lee \(2025\)](#) link occupational exposure scores to CPS data and, using a first-difference design, show that higher GenAI exposure is associated with reduced employment. More recently, [Lodefalk et al. \(2026\)](#) use Swedish job-posting and register data in an employer-level difference-in-differences design and find a widening age gradient in hiring: since the release of ChatGPT, employment of workers aged 22–25 in high-exposure occupations has fallen by about 5.5 percent relative to less-exposed occupations within the same employers, while workers over 50 have seen a slight increase. [Crane and Soto \(2026\)](#) focus on computer-programming occupations—among the most AI-exposed—and, using a within- versus between-industry decomposition of CPS data, find that coder employment growth is roughly 3 percentage points lower post-ChatGPT than pre-ChatGPT, even after controlling for industry-level shocks.¹⁰ By contrast, [Chandar \(2025\)](#) and [Financial Times \(2025\)](#) do not find systematic differences in employment patterns between more- and less-exposed occupations in CPS data. [Eckhardt and Goldschlag \(2025\)](#) compare unemployment patterns using five exposure measures, finding statistically significant differences for only two, with even those effects relatively small.

Our contribution to this literature is to move beyond occupation-level exposure indices and provide broad-based evidence using a more direct measure of firm-level adoption, identified through postings for explicit “GenAI integrator” roles. Moreover, unlike exposure-based measures, our adoption measure varies over time, allowing us to exploit variation in adoption timing with an event-study design.

¹⁰Motivated by this body of evidence, [Friebel et al. \(2026\)](#) model the dynamic implications of reduced junior hiring for the short- and long-run structure of firms’ workforce. Relatedly, [Ide \(2025\)](#) develop an overlapping-generations model in which AI-driven entry-level automation weakens the intergenerational transmission of tacit knowledge from experts to novices, potentially reducing long-run growth.

comparing adopters to non-adopters. Firm-by-time fixed effects γ_{it} absorb any shocks or trajectories specific to a given firm in a given period. The industry-by-seniority-by-time fixed effects $\xi_{p(i)st}$ control for broad sectoral trends that differentially affect juniors and seniors over time, ensuring that our results are not driven by sector-wide shifts in junior versus senior employment.

Figure 5 presents the estimated coefficients β_j from Equation 11. Estimation begins in 2018Q1 due to computational constraints. The coefficients remain essentially flat through 2022Q4, aside from a brief dip in early 2021.²⁶ Beginning in 2023Q1, the coefficients drop sharply, reaching roughly a 9 percent decrease after six quarters. This break—coinciding with the rapid diffusion of GenAI—provides suggestive evidence that adoption is associated with increasingly seniority-biased patterns, reducing junior employment relative to senior employment within firms.²⁷

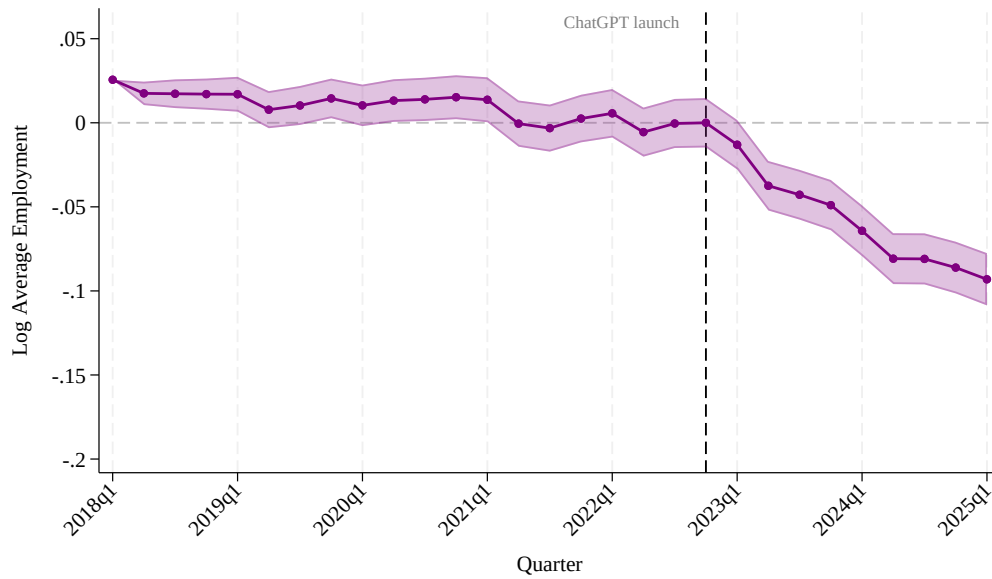


Figure 5: Triple-Difference Estimates: Juniors vs. Seniors

Notes: The figure presents the estimated triple-difference coefficients β_j from Equation 11, capturing how junior employment evolves relative to senior employment within the same firm and period, comparing adopters to non-adopters. Standard errors are clustered at the firm level.

The β_j coefficients can be interpreted as reflecting the differential impact of GenAI

²⁶The patterns shown in Panels 4a and 4b indicate that this temporary decline likely reflects a brief acceleration in senior employment among adopting firms in 2021, rather than a decrease in junior employment.

²⁷In Supplemental Appendix A.18, we present triple-difference estimates excluding the industry-by-seniority-by-time fixed effects. The results are very similar to those in Figure 5.

Table 4: Hiring, Separations, and Promotions

	Hiring	Separation	Promotion (rate)
Treat $\times$ Post	−4.006*** (0.221)	−1.089*** (0.163)	0.033** (0.014)
Pre-period mean	5.059	4.154	0.822
Observations	7,941,179	7,941,179	7,690,581
Clusters (firms)		280,866	

Notes: This table reports the estimated β from Equation 15. Estimation begins in 2018 to exclude early-sample trends. Standard errors clustered by firm in parentheses.

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

6 Conclusion

This paper provides early, large-scale evidence that the diffusion of GenAI since 2023 is associated with *seniority-biased* employment effects within firms. Using résumé and job-posting data linked to roughly 281,000 U.S. firms, together with a firm-level adoption measure based on “GenAI integrator” vacancies, we document a decline in junior employment relative to senior employment at adopting firms compared with non-adopting firms. This decline is concentrated in high-exposure occupations: within adopting firms, junior employment in high-exposure occupations contracts relative to low-exposure occupations after 2022, while the corresponding senior series continues to rise. Task-level evidence from job postings points to the same mechanism: GenAI-exposed tasks become less likely to appear in junior task bundles at adopting firms. Finally, decomposing employment flows shows that the decline is driven primarily by slower hiring rather than increased separations.

These findings should be interpreted with caution. Adopting firms differ systematically in size, workforce composition, and industry. Although our research designs absorb many observable and unobservable differences across firms, sectors, and time, unobserved confounding factors may remain. We therefore examine four prominent alternative explanations: monetary tightening, post-pandemic hiring dynamics, the expansion of remote work, and labor-supply shocks. These tests support our interpretation, but they cannot rule out all alternative explanations. In addition, our adoption measure captures deliberate organizational integration of GenAI but may miss informal or “silent” adop-

tion within firms. Finally, because GenAI is still in the early stages of diffusion, longer-run adjustments in training, task allocation, wages, and internal career ladders may either attenuate or amplify the effects documented here.

Even with these caveats, the results suggest that GenAI adoption may be shifting labor demand away from entry-level tasks and narrowing the lower rungs of internal career ladders. Because early-career jobs are central to skill formation and lifetime earnings growth, such changes may have consequences beyond short-run employment. Whether these patterns persist, and whether firms respond by redesigning training, tasks, or career progression, remains an important question for future research.

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